Unit 6, Lesson 1: Dilations

Benchmark

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

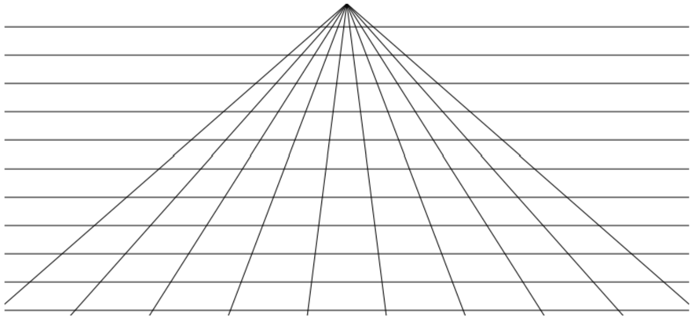
Learning Targets

- ☐ I can write an algebraic description using coordinate notation of a dilation on a coordinate plane.
- ☐ I can describe a dilation by giving the scale factor and center of dilation.

6.1.1 Warm-Up: What Is Your Perspective?

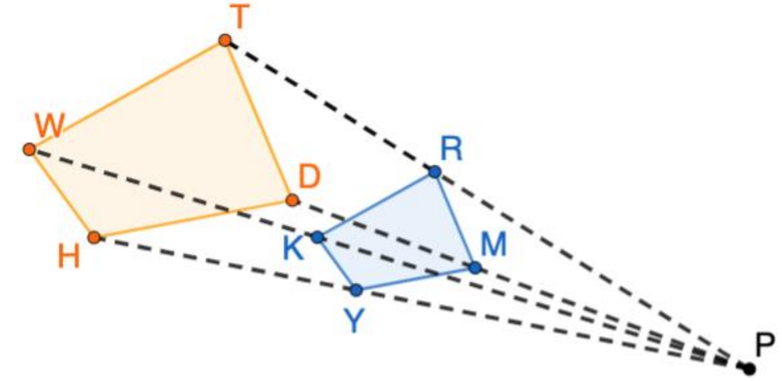
Point perspective is a common method of drawing a three-dimensional object onto a two-dimensional plane. The artist chooses one point from which everything in the drawing is set. It was first discovered during the Renaissance. An example of a painting that uses point perspective is Vincent van Gogh's *Vincent's Bedroom in Arles*.

1. Use the grid to sketch a rectangle that is close to the point of perspective and a rectangle that is farther away from the point of perspective.



6.1.2 Refresher: Dilations

1. $KRMY$ was created by dilating $WTDH$. The ratio of WT to KR is $\frac{5}{3}$.



   a. Use the words in the bank below to complete the statements that follow.

center of dilation	image
pre-image	scale factor

Point P is the ______________________.

$KRMY$ is the ______________________.

$WTDH$ is the ______________________.

The ______________________ is $\frac{3}{5}$.

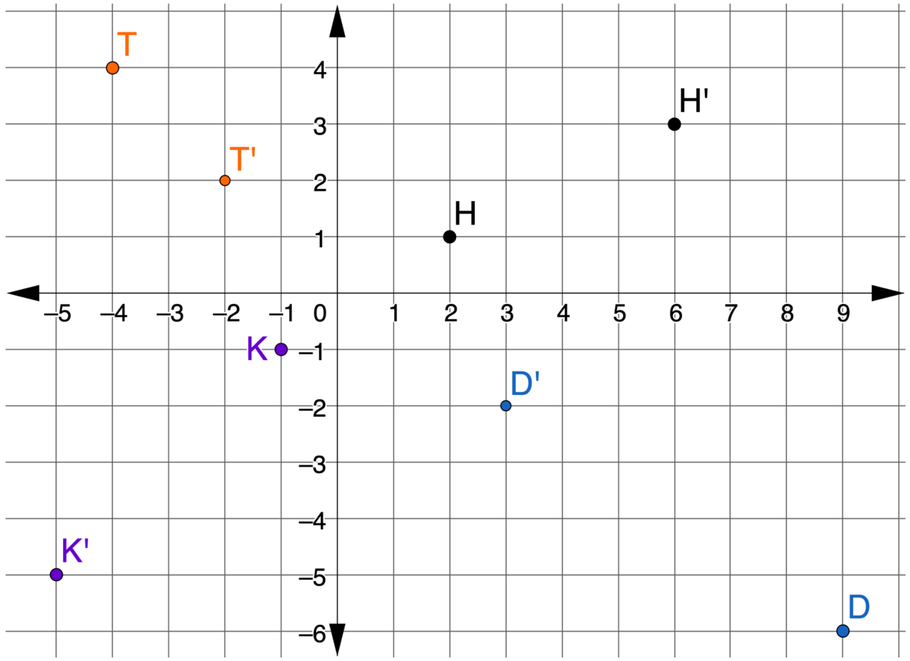
   b. Use the diagram to determine each.

What is the ratio of $\frac{WH}{KY}$?

What is the ratio of $\frac{PR}{RT}$?

6.1.3 Collaboration: Algebraic Description of a Dilation

1. Points D, H, K, and T are shown on the coordinate grid. Each point has been dilated using $(0, 0)$ as the center of dilation.



   a. Complete the table.

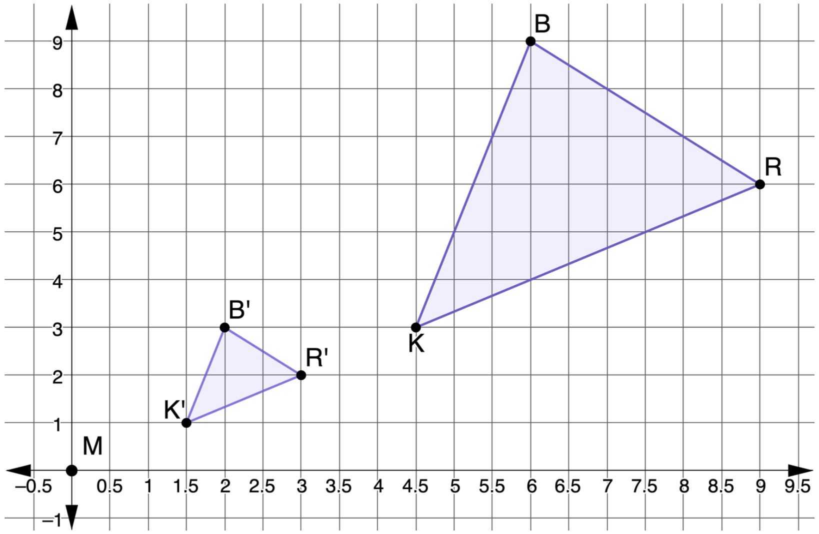
Point	Ordered Pair	Ordered Pair of the Dilation	Scale Factor
D			
H			
K			
T			

In the coordinate plane, the algebraic representation of a dilation centered at the origin is $(x, y) \rightarrow (kx, ky)$, where k is the scale factor.

b. Write the algebraic representation for each point.

Point	Algebraic Representation of Dilation
D	
H	
K	
T	

2. $\Delta R'B'K'$ is the result of dilating ΔRBK using the origin as the center of dilation.



a. What is the scale factor of the dilation?

b. Write the algebraic description of the dilation.

c. Draw a line segment from point M that passes through B' and ends at B.

d. The table shows different relationships of the line segments MB, $B'B$, and MB'.

Line Segment	Slope	Distance
MB'	$\frac{3}{2}$	$\sqrt{2^2+3^2}=\sqrt{13}$
$B'B$	$\frac{6}{4}$	$\sqrt{6^2+4^2}=\sqrt{52}=2\sqrt{13}$
MB	$\frac{9}{6}$	$\sqrt{6^2+9^2}=\sqrt{117}=3\sqrt{13}$

e. Discuss with your partner how the scale factor could be determined using slope. Summarize your discussion.

f. Discuss with your partner how the scale factor could be determined using distance.

3. The table shows different relationships among the line segments BR, $B'R'$, KR and $K'R'$.

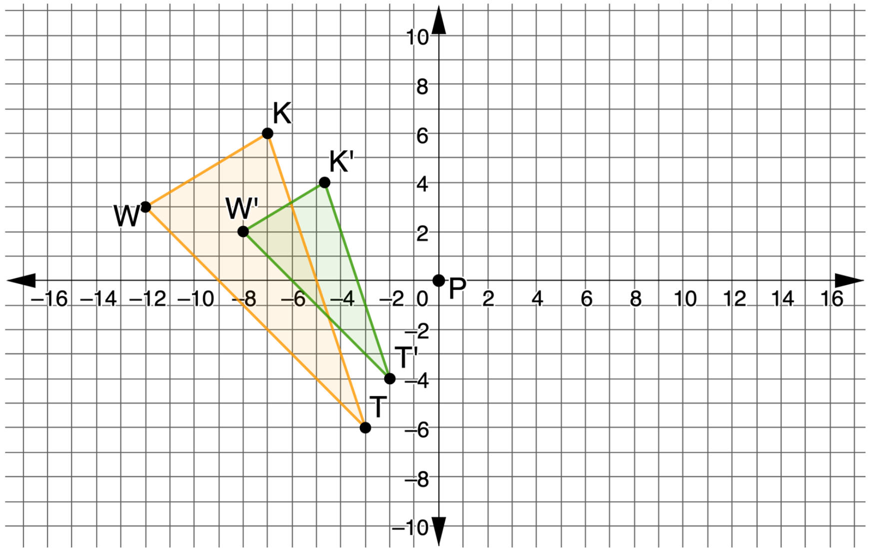
Line Segment	Slope	Distance
BR	$-\frac{3}{3}$	$\sqrt{3^2+3^2}=\sqrt{18}=3\sqrt{2}$
$B'R'$	$-\frac{1}{1}$	$\sqrt{1^2+1^2}=\sqrt{2}$
KR	$\frac{3}{4.5}$	$\sqrt{3^2+4.5^2}=\sqrt{29.25}$
$K'R'$	$\frac{1}{1.5}$	$\sqrt{1^2+1.5^2}=\sqrt{3.25}$

Discuss with your partner how the scale factor is also found in the relationships between the slopes of BR and $B'R'$, and KR and $K'R'$. Write a summary of the relationships.

6.1.4 Guided Instruction: Describing Dilations

1. $\Delta K'M'T'$ is the result of dilating ΔKMT using the origin as the center of dilation. The vertices of ΔKMT are located at $K(-8,-6)$, $M(6,7)$, and $T(0,4)$. The vertices of $\Delta K'M'T'$ are located at $K'(-12,-9)$, $M'(9,10.5)$, and $T'(0,6)$. Write the algebraic description of the dilation.

2. Triangles TWK and $T'W'K'$ are shown on the coordinate grid, where $\Delta T'W'K'$ is a dilation of ΔTWK and point P is the center of dilation.

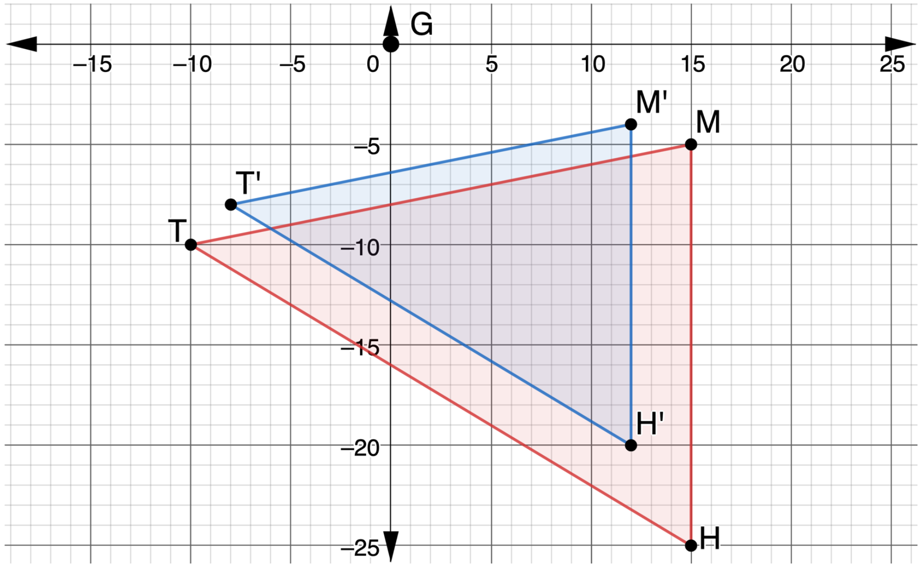


a. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.

b. What is the scale factor of the dilation?

c. Write the algebraic description of the dilation.

3. Triangles THM and $T'H'M'$ are shown on the coordinate grid, where $\Delta T'H'M''$ is a dilation of ΔTHM.

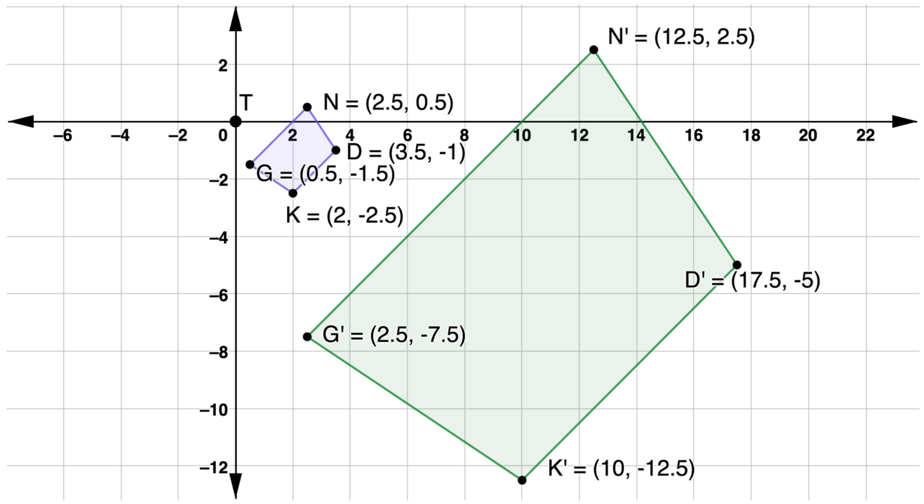


Write a description of the dilation by giving the center of dilation and the scale factor.

6.1.5 Guided Practice: Describing Dilations

1. $\overline{M'T'}$ has coordinates $M'(-16,20)$ and $T'(4,-8)$, and is the result of the dilation of $\overline{MT}$ centered at the origin. The coordinates of $\overline{MT}$ are $M(-4,5)$ and $T(1,-2)$. Write the algebraic description of the dilation.

2. $NDKG$ and $N'D'K'G'$ are shown on the coordinate grid, where $N'D'K'G'$ is a dilation of $NDKG$.



Write a description of the dilation by giving the center of dilation and the scale factor.

6.1.6 Wrap-Up: Cool Down

1. Explain how dilations are different from rigid transformations, in your own words.

Unit 6, Lesson 2: Sequences of Transformations That Include Dilations

Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

Learning Targets

- ☐ I can describe a sequence of transformations that includes a dilation.
- ☐ I can represent a sequence of transformations using coordinate notation with algebraic descriptors.